

Choosing the Right Kind of Intelligence: Humans, Rules, ML and Generative AI

Learn to match a problem to the cheapest system that can actually solve it, and to recognise when AI is the wrong tool.

For engineers and technical leads deciding where AI belongs in a system and where it does not · 26 July 2026

The one- or two-page version: what matters, the topics it covers, and every term defined. Read the full lesson for the explanations and worked examples.

[Watch the source video](#)[Read the full lesson](#)[Read the condensed version](#)

What matters

- ✓ Four options — humans, rules, ML, generative AI — are distinct tools, not rungs on a ladder.
- ✓ Humans are the right choice for high-stakes, ambiguous, accountability-bearing decisions, and they do not scale.
- ✓ Rules are the right choice when logic is known, explicit and stable, and when errors are unacceptable.
- ✓ Rules break when conditions change faster than you can update them, or when the logic grows too complex to write down.
- ✓ Machine learning fits when patterns exist in structured data but are not obvious enough to encode by hand.
- ✓ ML is not fire-and-forget: drift monitoring and retraining are permanent operating costs.
- ✓ Generative AI fits unstructured input, interpretation and synthesis — where flexibility matters more than exactness.
- ✓ The deciding question for generative AI is how much non-determinism you can tolerate, and the best systems combine all four.

What it covers

01 Four kinds of intelligence, not a ladder

0:00

Humans, rules, machine learning and generative AI are distinct options with distinct roles. Choosing among them is an ordinary engineering trade-off.

02 Humans: judgement, ambiguity and accountability

1:27

People handle incomplete and conflicting information, apply ethics and context, and can be held responsible. They are also expensive and do not scale.

03 Rules and code: deterministic logic at scale

3:01

When the logic is known, stable and must not be wrong, ordinary code is fast, cheap, testable and completely interpretable.

04 Where rules break down

3:44

Rules fail when conditions change constantly or the logic becomes too complex to state. Fraud detection is the standard example.

05 Machine learning: patterns in structured data

4:29

Learn the distinguishing pattern from examples instead of stating it. Right when patterns exist but are not obvious enough to encode by hand.

06 **What machine learning costs to keep running** 5:13

Models scale well and behave predictably, and they need monitoring, retraining and an honest account of what they can explain.

07 **Generative AI: unstructured input, flexible output** 5:53

Right when inputs are text or media, tasks need interpretation or transformation, and flexibility matters more than precision.

08 **The real question: how much non-determinism can you tolerate?** 7:21

Generative systems do not guarantee correctness across runs, which makes them hard to test and costly at scale. That tolerance is the deciding criterion.

09 **Hybrid systems, and the heuristic** 8:11

The best systems use all four in combination. 'Make everything agents' was never meant literally.

Glossary

Deterministic system — One where identical input always produces identical output. Exhaustively testable, and its behaviour is fixed once verified.

Machine learning — Deriving a predictive function from labelled examples rather than stating it. Suits structured data with patterns too complex to encode by hand.

Data drift — A change in the distribution of incoming features relative to training. Detectable by comparing distributions.

Population Stability Index (PSI) — A measure of how much a feature's distribution has shifted from a baseline. Commonly read as stable below 0.1 and needing retraining above 0.2.

Golden set — A fixed collection of inputs with known-good outputs, scored repeatedly to catch regressions in a system you cannot assert on exactly.

Feature attribution — Identifying which inputs most influenced a prediction. Explains the model's behaviour, never the underlying causality.

Rule-based system — Explicit conditional logic written by a person. Fast, cheap, interpretable and correct only for the cases anticipated.

Model drift — Gradual degradation in accuracy as the world diverges from the training data. Silent: no error, no code change.

Concept drift — A change in the relationship between inputs and outcomes. Harder to detect, because it requires outcomes that often arrive late.

Non-determinism — The property that identical input may produce different output across runs. The defining cost of generative systems.

LLM as judge — Using a second model to score an output against a rubric, where the output is too open-ended for a programmatic check.

Hybrid system — An architecture combining humans, rules, ML and generative AI, each applied to the part of the problem it fits.

Explanations are AI-generated. Check anything you intend to rely on.